Alireza Moradi

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Research Interests

Renewable Forecasting, Solar Forecasting, Uncertainty quantification, Reliability, Conformal Prediction

Education

Georgia Institute of Technology, Atlanta, GA
Doctor of Philosophy (Ph.D.) in Machine Learning — 2025 – Present (Expected 2029)

- Advisor: Prof. Pascal Van Hentenryck. Graduate Research Assistant, *AI4OPT*.
- GPA: 3.72 / 4.00

Georgia Institute of Technology, Atlanta, GA
M.S. in Computer Science — 2025 – 2026

- GPA: 3.72 / 4.00

Sharif University of Technology, Tehran, Iran
B.S. in Industrial Engineering, *Minor in Computer Science* — 2019 – 2024

- Thesis: time series modeling for power system forecasting.
- GPA: 3.76 / 4.00

Publications

Published / Accepted

[1] **A. Moradi** and N. Sedghi, "Forecasting Electricity Demand and Managing Supply Risk for Sustainable Energy Policies," *Reliability Engineering & System Safety*, accepted, 2026. DOI: 10.1016/j.ress.2026.113305.

[2] M. Ghasemloo and **A. Moradi**, "Informed Forecasting: Leveraging Auxiliary Knowledge to Boost LLM Performance on Time Series Forecasting," *PLOS ONE*, accepted, 2026. DOI: 10.1371/journal.pone.0358970.

[3] A. Jezghani, B. Crawford, R. Barrass, M. Butros, **A. Moradi**, and P. Van Hentenryck, "Lowering the Barrier: Leveraging JupyterHub for K–12 AI and Data Science Outreach," *SC Workshop on Best Practices for HPC Training and Education*, accepted, 2026.

[4] **A. Moradi**, M. Tanneau, R. Zandehshahvar, and P. Van Hentenryck, "Enhanced Renewable Energy Forecasting and Operations through Probabilistic Forecast Aggregation," *Proceedings of the IISE Annual Conference and Expo*, 2025. DOI: 10.21872/annual2025_9066.

Under Review

[1] **A. Moradi**, T. Bruys, P. Van Hentenryck, and R. Zandehshahvar, "CoCoNet: Convex Conformal Network for Multi-Dimensional Uncertainty Regions," under review, 2026.

[2] **A. Moradi**, M. Tanneau, R. Zandehshahvar, and P. Van Hentenryck, "Enhanced Renewable Energy Forecasting using Context-Aware Conformal Prediction," under review, 2026.

[3] T. Bruys, S. Cheng, **A. Moradi**, R. Zandehshahvar, and P. Van Hentenryck, "Toward Reliable Generative AI with Conformal Prediction: A Survey," under review, 2026.

[4] **A. Moradi** et al., "Copula-Based Aggregation and Context-Aware Conformal Prediction for Reliable Renewable Energy Forecasting," under review, 2026.

Research Experience

Graduate Research Assistant, AI4OPT — Georgia Institute of Technology, Atlanta, GA — Sep 2024 – Present

- Research on probabilistic forecasting, conformal prediction, and uncertainty quantification for large-scale energy systems, advised by Prof. Pascal Van Hentenryck.

- Designed a context-aware conformal prediction method for renewable energy forecasting on large-scale systems (SPP, MISO, ERCOT).

Undergraduate Research Assistant — Sharif University of Technology, Tehran, Iran Sep 2022 – Jun 2024

- Developed probabilistic electricity-demand forecasting models using quantile regression and machine learning, with supply-risk analysis (published in *Reliability Engineering & System Safety*).

Industry Collaborations

Réseau de Transport d'Électricité (RTE), France — TS4PS project (PSERC)

- Led the reconstruction of multi-year, 5-minute nodal active-power injection time series for the RTE7000 French transmission grid (∼7,000 buses) by capacity-proportional disaggregation of public regional data (eco2mix, ENTSO-E).
- Released the open-source pipeline on GitHub and the dataset (∼12.8B rows, 2021–2023) on Hugging Face.

U.S. Vertically Integrated Utility

- Developed a calibrated probabilistic aggregation framework that turns site-level solar forecasts from heterogeneous third-party providers into reliable fleet-level forecasts, where system-level models cannot be trained or maintained.
- Combined copula-based modeling of cross-site dependence with context-aware conformal prediction (CACP) to correct aggregation-induced miscalibration, reaching near-nominal coverage with significantly sharper intervals than uncalibrated aggregation on large-scale solar data.

Teaching Experience

Instructor, AI for Time series Forecasting — Georgia Institute of Technology Spring 2026 – Present

- Mentoring an undergraduate team building a generative AI framework that combines market data and financial news for time-series forecasting.

Instructor / Head Instructor, Seth Bonder Camp — AI4OPT, Georgia Tech Summer 2025 – Summer 2026

- Taught (2025) and led the instructional team (2026) of an AI and data science outreach camp for high school students.

Teaching Assistant — University of Calgary (remote) Jan 2024 – May 2024

- Systems Analysis.

Teaching Assistant — Sharif University of Technology Sep 2020 – Jun 2022

- Simulation, Operations Research, Transportation Planning, and Python Programming.

Technical Skills

- **Programming:** Python (PyTorch, scikit-learn, pandas), Julia, SQL, LaTeX.
- **Methods:** Conformal prediction, quantile regression, deep learning for time series, stochastic optimization, simulation (discrete-event, agent-based, system dynamics).
- **Tools:** Git, Linux/HPC (Slurm, GPU clusters), Gurobi.